

Spotting the weird ones

Rob J Hyndman

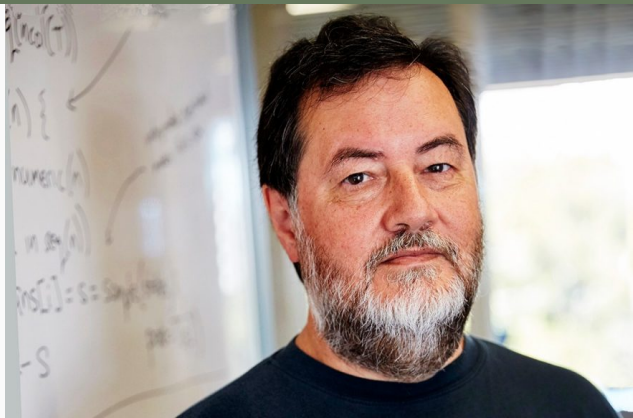
6 August 2026

robjhyndman.com/ihaka2026



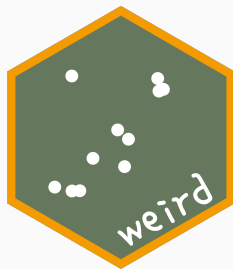
Thanks Ross!

- Ross Ihaka is my academic great uncle
- 1987: I first used S
- 1988: I switched to S-PLUS
- 1994: I heard Ross and Robert give a talk on R at ASC
- 2000: I switched to R
- 2006: My first CRAN package
- 2026: Coauthor of 63 R packages
- 2023–2026: Executive Editor, *R Journal*



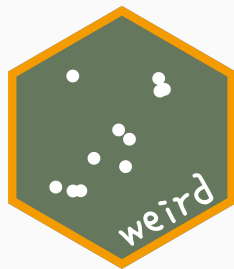
weird R package

- Functions and data sets required for *That's Weird: Anomaly Detection Using R*
[OTexts.com/weird](https://otexts.com/weird)
- All packages needed to run the examples are also loaded



weird R package

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```
library(weird)
```

```
— Attaching core weird packages — weird 3.0.0 —  
✓ distributional 0.8.1  
✓ ggplot2        4.0.3  
✓ dplyr          1.2.1  
— Conflicts — weird_conflicts —  
✖ dplyr::filter() masks stats::filter()  
✖ dplyr::lag()    masks stats::lag()
```

Outline

- 1 What is an anomaly?
- 2 Wine quality and prices
- 3 Approximate extreme surprisal probabilities
- 4 Cricket not-outs
- 5 Old Faithful eruptions
- 6 French mortality rates

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What do we mean by “weird”?

an observation (or a subset of observations) which appears to be inconsistent with the remainder of that set of data.

(Barnett & Lewis, 1978)

What do we mean by “weird”?

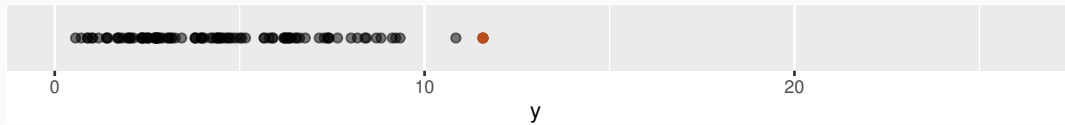
an observation (or a subset of observations) which appears to be inconsistent with the remainder of that set of data.

(Barnett & Lewis, 1978)

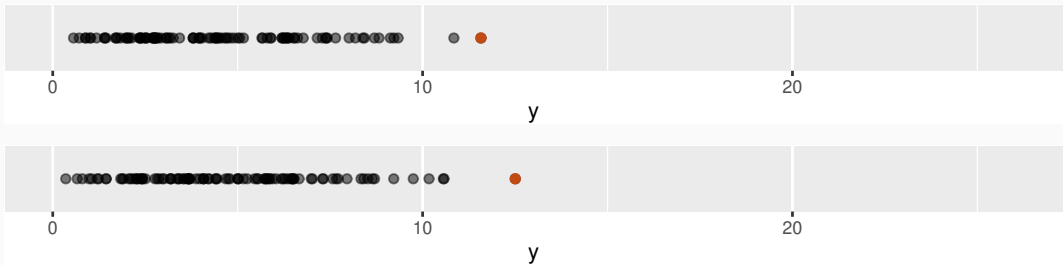
an observation which deviates so much from other observations as to arouse suspicion it was generated by a different mechanism.

(Hawkins, 1980)

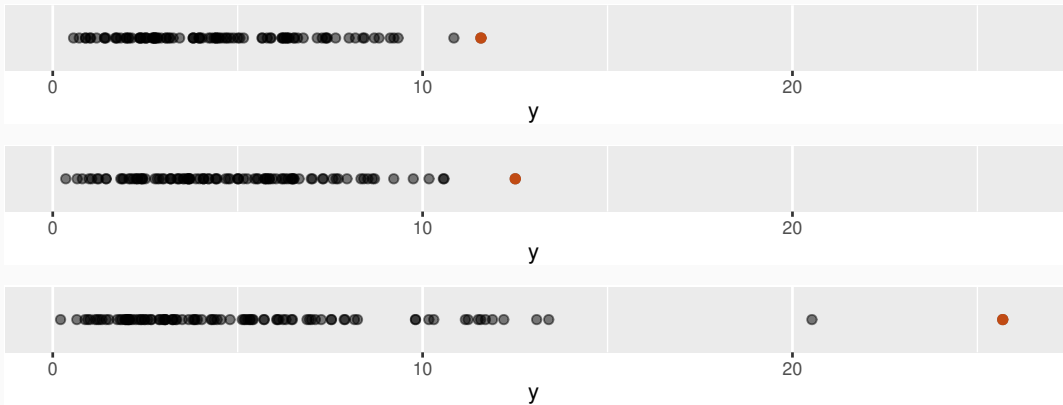
Is this an anomaly?



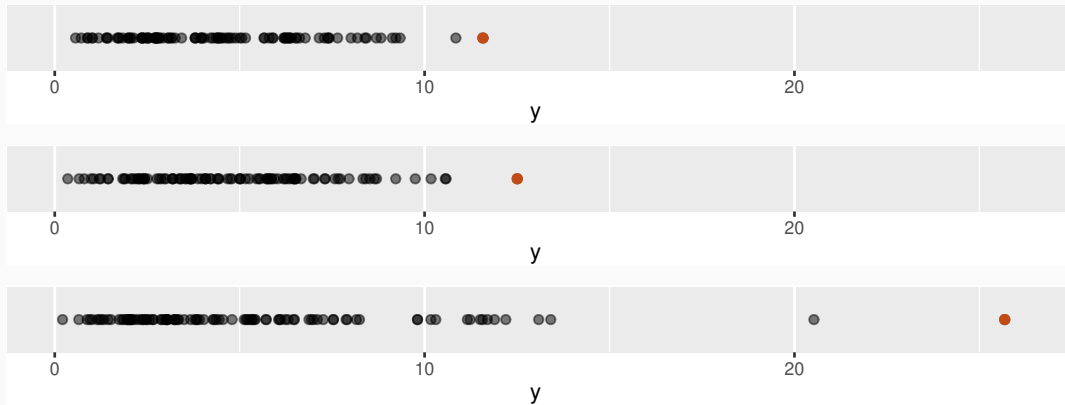
Is this an anomaly?



Is this an anomaly?

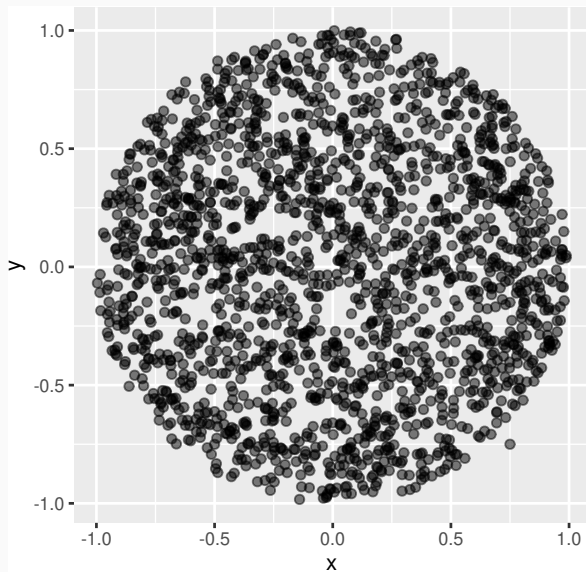


Is this an anomaly?

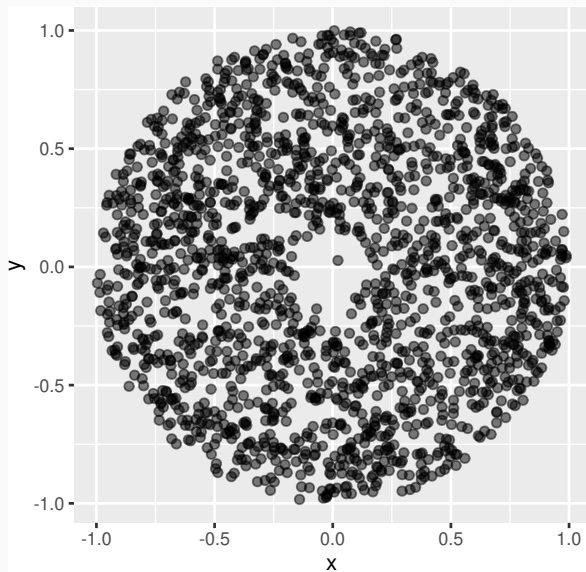


All points were randomly generated from the **same** χ^2_5 distribution.

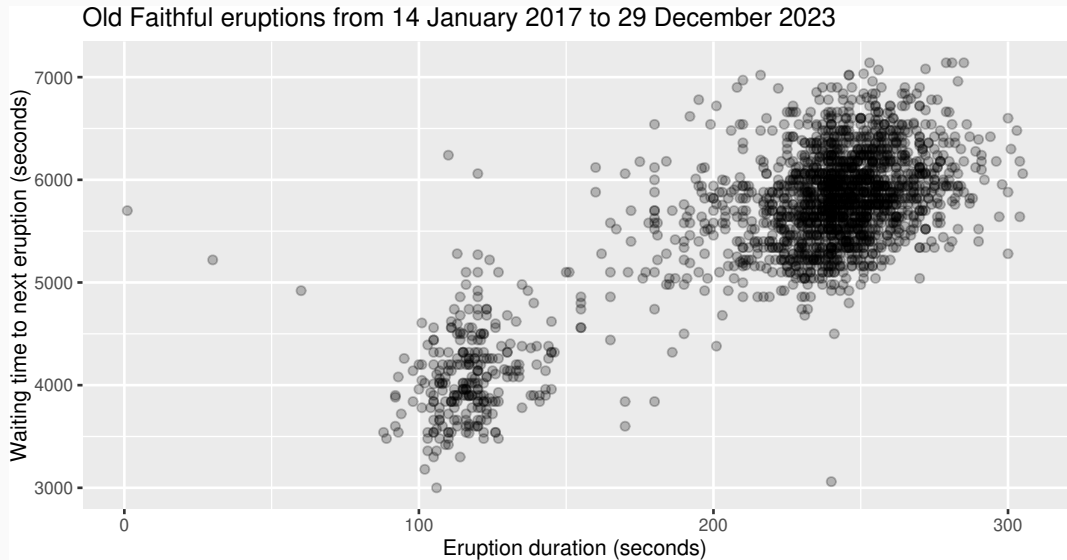
Spot the anomaly



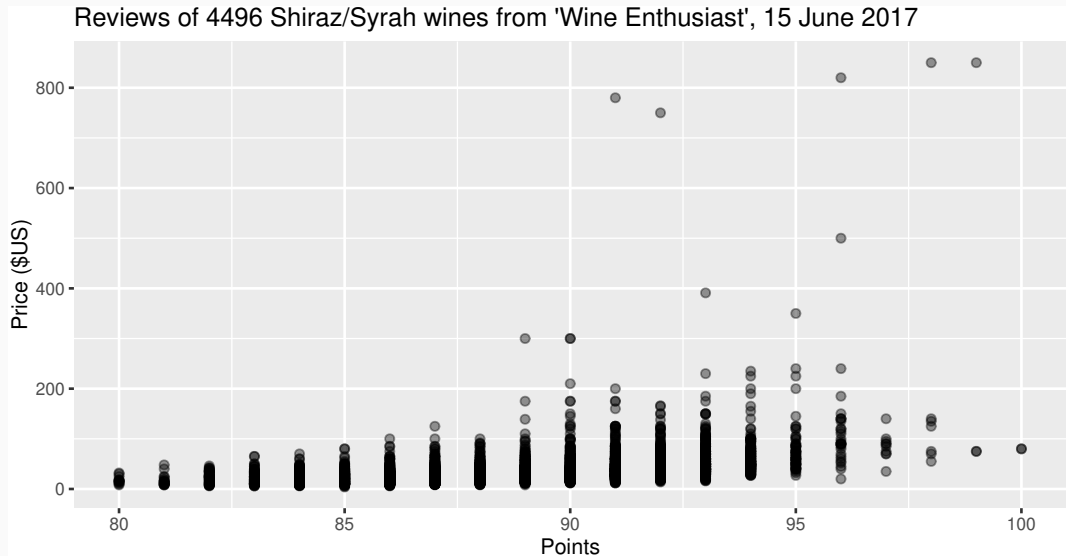
Spot the anomaly



Are there any anomalies here?



Are there any anomalies here?



Surprisal probabilities

Surprisal probability of observation y_i under model f

The probability of an observation that is at least as unlikely:

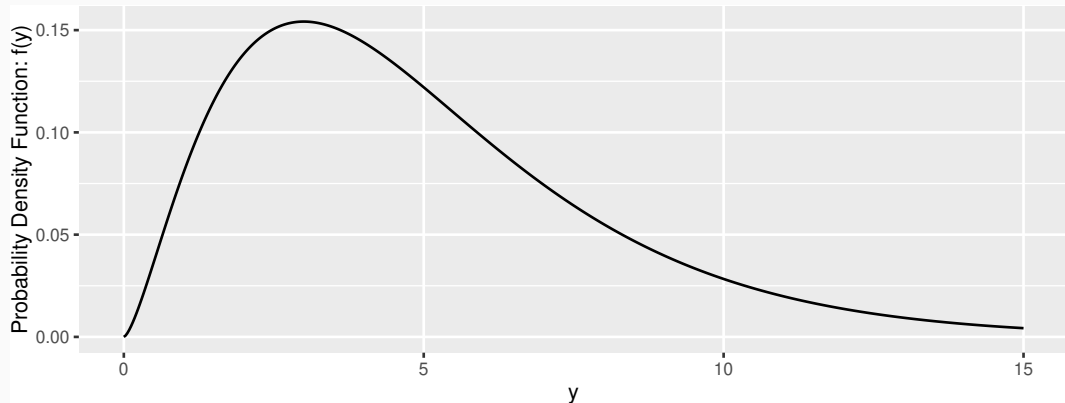
$$p_i = \mathbb{P}(f(Y) \leq f(y_i)) \quad \text{where } Y \text{ has density } f$$

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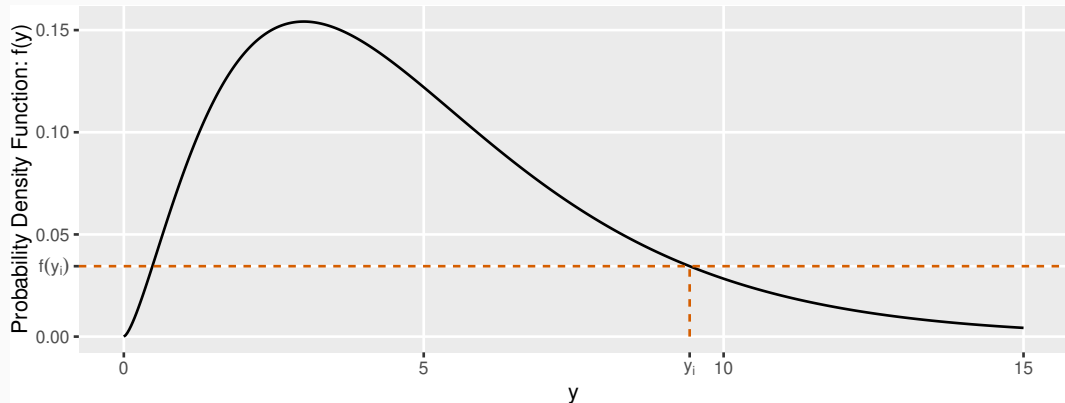


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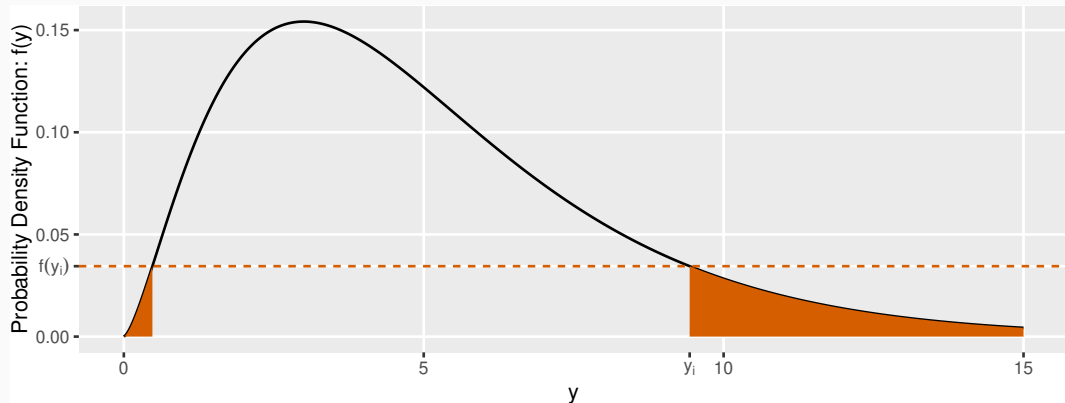


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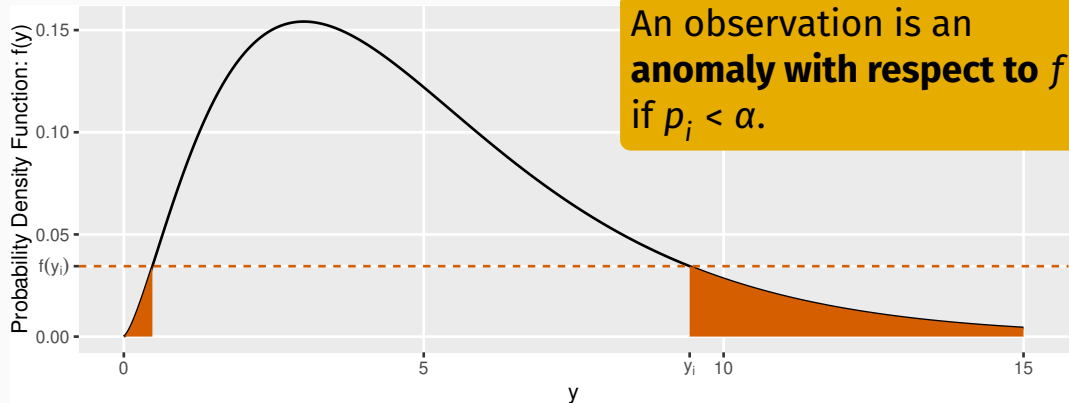


Surprisal probabilities

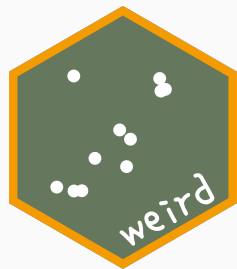
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weird R package



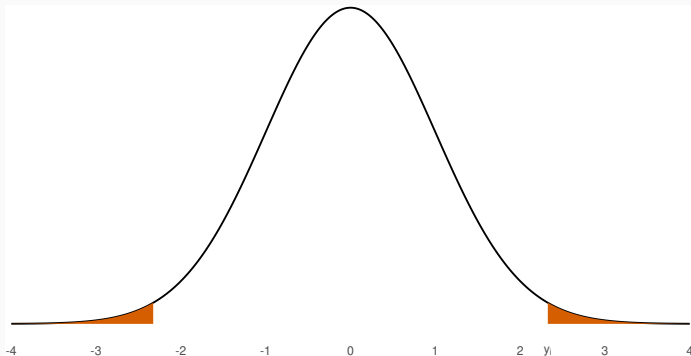
- `surprisals_prob( )` is an S3 generic that computes surprisal probabilities for any data object using a user-specified density function, or a kernel density estimate. It also works with linear models, GLMs, and GAMs.
- Distributions can be specified as distributional objects from the distributional package.

weird R package

```
N01 <- dist_normal(mean = 0, sd = 1)
tibble(
  y = generate(N01, times = 1000)[[1]],
  p = surprisals_prob(y, distribution = N01)
) |>
  arrange(p)
```

A tibble: 1,000 x 2

	y	p
	<dbl>	<dbl>
1	3.42	0.000634
2	3.14	0.00167
3	-3.05	0.00230
4	-2.96	0.00304
5	-2.89	0.00390
6	2.81	0.00488
7	2.64	0.00832
8	-2.60	0.00923
9	2.56	0.0104

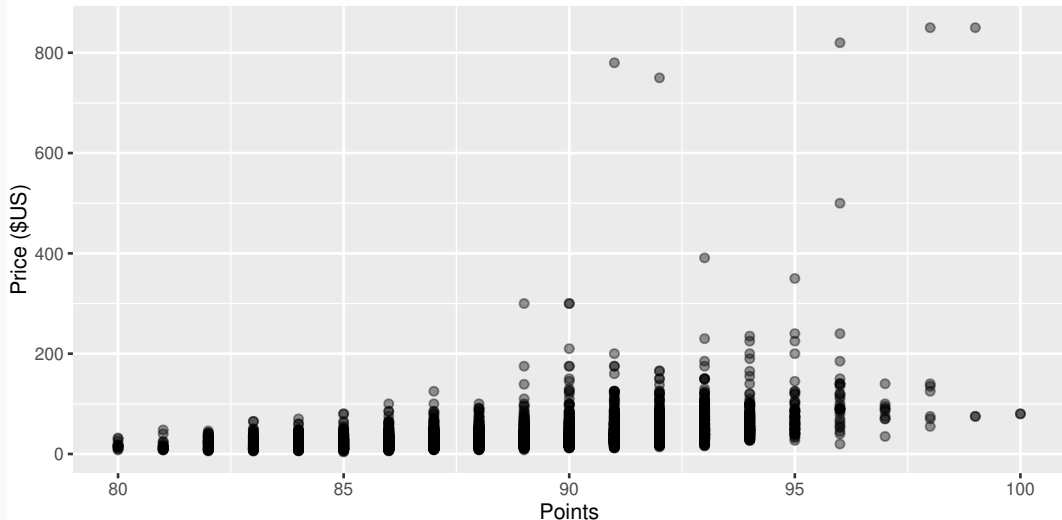


Outline

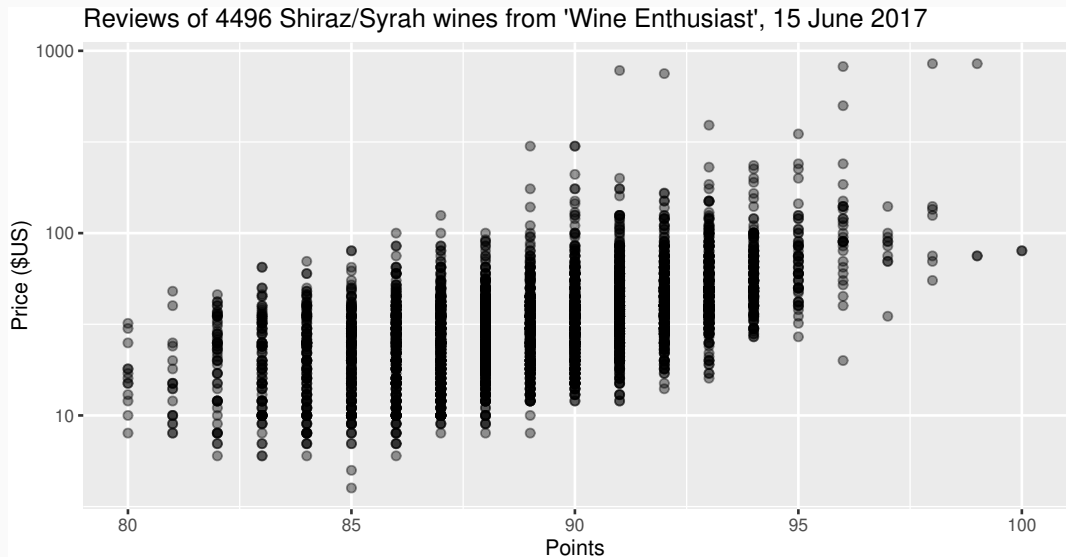
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Wine quality and prices

Reviews of 4496 Shiraz/Syrah wines from 'Wine Enthusiast', 15 June 2017



Wine quality and prices

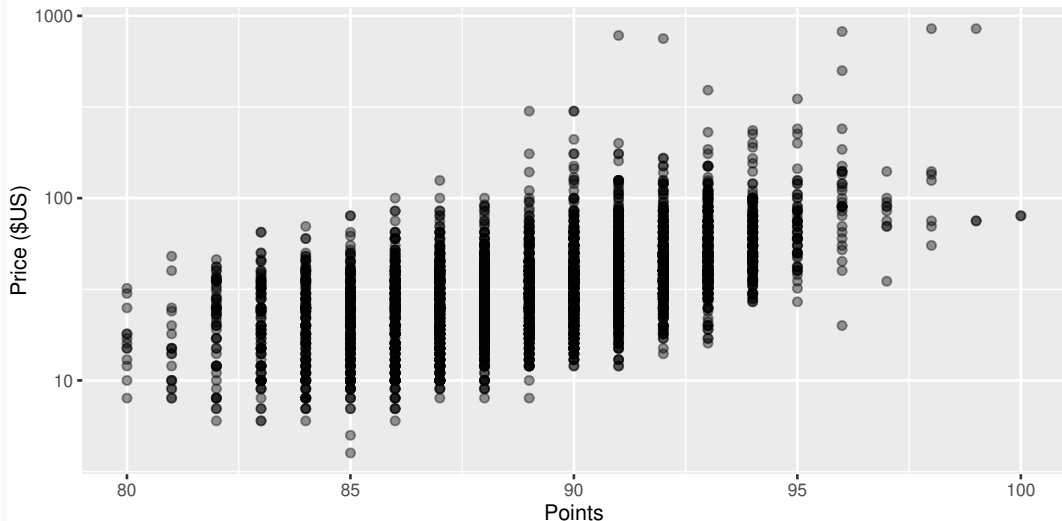


Wine quality and prices

Statistical model

$$\log(\text{Price}) \sim N(a + b\text{Points}, \sigma^2)$$

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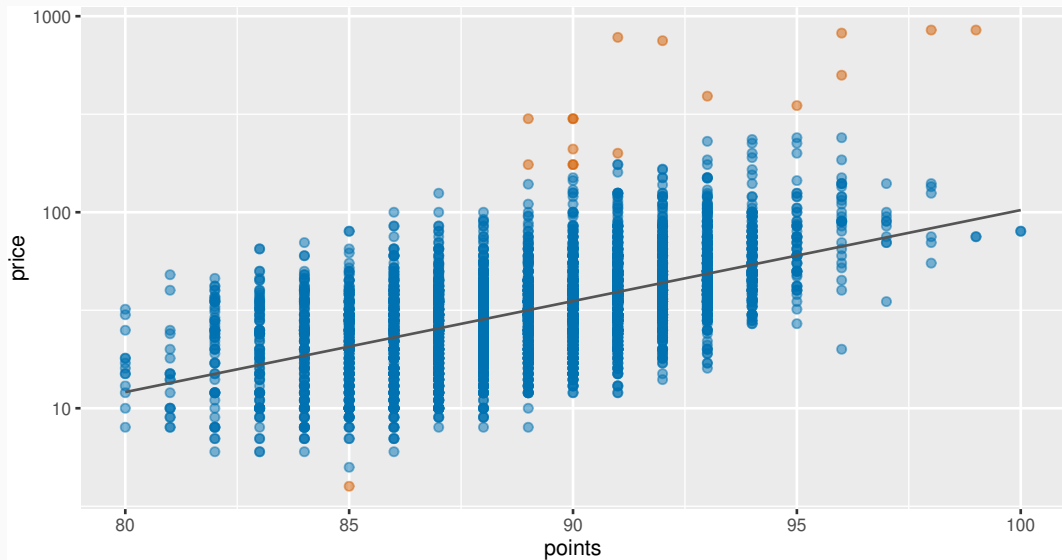
```
wine_lm <- lm(log(price) ~ points, data = wine_reviews)
wine_reviews <- wine_reviews |>
  mutate(p = surprisals_prob(wine_lm, loo = TRUE))
wine_reviews |> arrange(p)
```

A tibble: 4,496 x 7

	country	state	winery	year	points	price	p
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Australia	South Australia	Henschke	2009	91	780	5.77e-10
2	US	California	Law	2013	92	750	3.82e- 9
3	Australia	South Australia	Henschke	2010	96	820	2.12e- 7
4	Australia	South Australia	Penfolds	2008	98	850	1.43e- 6
5	Italy	Tuscany	Tua Rita	2011	89	300	3.30e- 6
6	Australia	South Australia	Penfolds	2010	99	850	4.24e- 6
7	Italy	Tuscany	Tua Rita	2013	90	300	9.46e- 6
8	Italy	Tuscany	Tua Rita	2012	90	300	9.46e- 6
9	France	Rhône Valley	Domaine Jean-Michel Ge~	2013	93	391	1.60e- 5
10	Australia	South Australia	Penfolds	2004	96	500	3.18e- 5

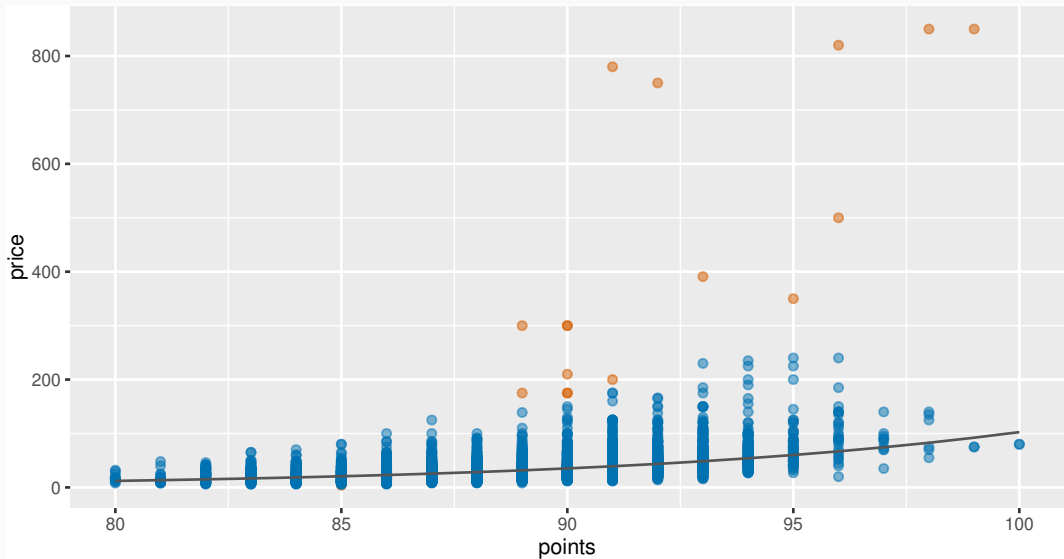
Wine quality and prices

$p_i < 0.001$



Wine quality and prices

$p_i < 0.001$



Weirdly low priced wines

```
wine_reviews |>  
  filter(log(price) < fitted(wine_lm)) |>  
  arrange(p, points, price)
```

```
# A tibble: 2,307 x 7
```

	country	state	winery	year	points	price	p
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Spain	Central Spain	Felix Solis	2013	85	4	7.00e-4
2	Australia	Australia Other	Banrock Station	2006	85	5	3.42e-3
3	France	Languedoc-Roussillon	Domaine Borie de M~	1999	89	8	4.52e-3
4	US	Washington	Two Vines	2012	86	6	5.58e-3
5	US	Washington	StoneCap	2010	88	8	8.84e-3
6	US	Washington	Rulo	2007	96	20	1.27e-2
7	Australia	Australia Other	Yellow Tail	2001	86	7	1.42e-2
8	France	Languedoc-Roussillon	Calvet	1998	86	7	1.42e-2
9	Chile	Central Valley	Morandé	2002	86	7	1.42e-2
10	India	Nashik	Sula	2012	91	12	1.46e-2

```
# i 2,297 more rows
```

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- **Surprisal**: $s_i = -\log f(y_i)$. Anomalies are extreme values in surprisal space.

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- We can use **extreme value theory** when f is wrong: the upper tail of the distribution of s_i is approximately Generalized Pareto Distribution, even if f is incorrect

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- f can be assumed, estimated, conditional, ...
- **Surprisal**: $s_i = -\log f(y_i)$. Anomalies are extreme values in surprisal space.
- We can use **extreme value theory** when f is wrong: the upper tail of the distribution of s_i is approximately Generalized Pareto Distribution, even if f is incorrect
- So $p_i = \mathbb{P}(-\log f(Y) \geq s_i) \approx 1 - F_{GPD}(s_i)$

Wine quality and prices

```
wine_reviews <- wine_reviews |>
  mutate(p = surprisals_prob(wine_lm, loo = TRUE, approximation = "gpd"))
wine_reviews |> arrange(p)
```

A tibble: 4,496 x 7

	country	state	winery	year	points	price	p
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Australia	South Australia	Henschke	2009	91	780	1.03e-4
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5	Italy	Tuscany	Tua Rita	2011	89	300	5.56e-4
6	Australia	South Australia	Penfolds	2010	99	850	5.93e-4
7	Italy	Tuscany	Tua Rita	2013	90	300	7.37e-4
8	Italy	Tuscany	Tua Rita	2012	90	300	7.37e-4
9	France	Rhône Valley	Domaine Jean-Michel Ger~	2013	93	391	8.56e-4
10	Australia	South Australia	Penfolds	2004	96	500	1.05e-3

i 4,486 more rows

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Cricket not-outs

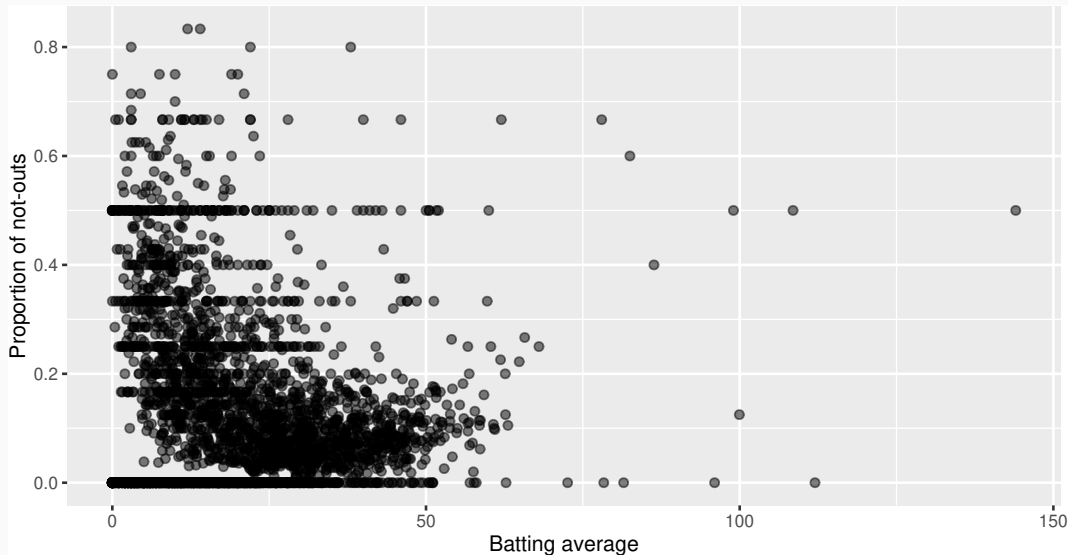
```
df_no <- cricket_batting |>
  filter(Innings > 0, Innings > NotOuts, Gender == "Men") |>
  mutate(prop_no = NotOuts / Innings)
df_no
```

A tibble: 3,144 x 16

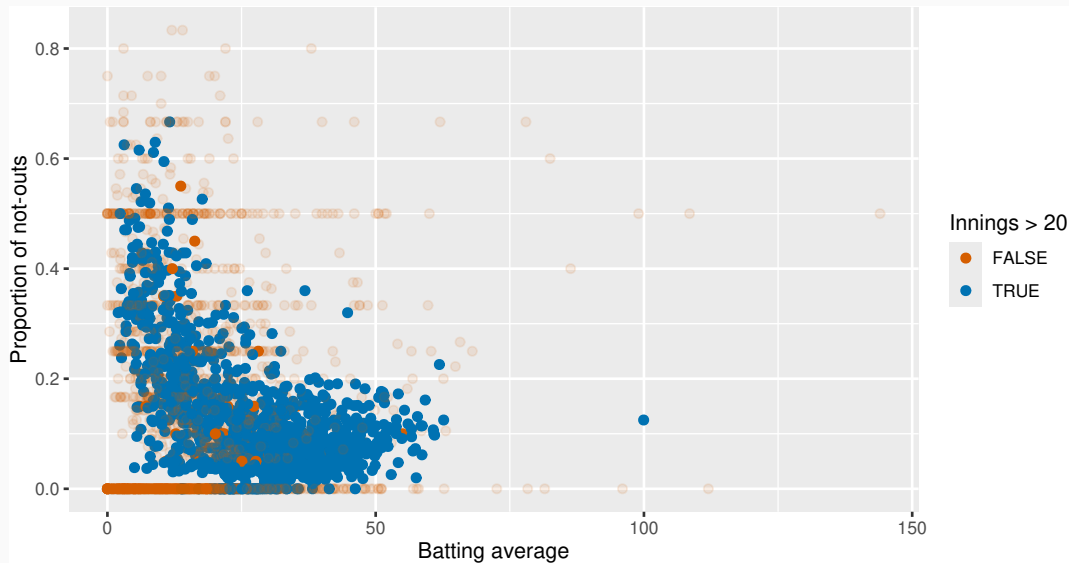
	Player	Country	Start	End	Matches	Innings	NotOuts	Runs	HighScore
	<chr>	<chr>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
1	BB Cooper	Australia	1877	1877	1	2	0	18	15
2	C Bannerman	Australia	1877	1879	3	6	2	239	165
3	DW Gregory	Australia	1877	1879	3	5	2	60	43
4	EJ Gregory	Australia	1877	1877	1	2	0	11	11
5	FR Spofforth	Australia	1877	1887	18	29	6	217	50
6	JM Blackham	Australia	1877	1894	35	62	11	800	74
7	JR Hodges	Australia	1877	1877	2	4	1	10	8
8	NFD Thomson	Australia	1877	1877	2	4	0	67	41
9	TJD Kelly	Australia	1877	1879	2	3	0	64	35
10	TK Kendall	Australia	1877	1877	2	4	1	39	17

i 3,134 more rows

Cricket not-outs



Cricket not-outs

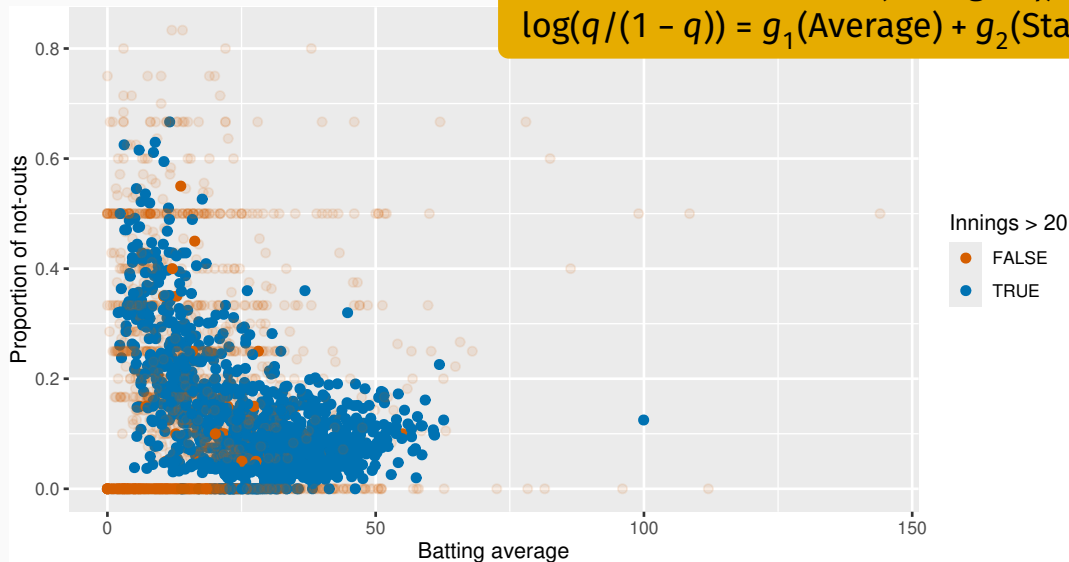


Cricket not-outs

Statistical model

NotOuts $\sim$ Binomial(Innings, q)

$\log(q/(1 - q)) = g_1(\text{Average}) + g_2(\text{Start})$



Cricket not-outs

```
fit_notouts <- mgcv::gam(  
  prop_no ~ s(Average) + s(Start),  
  data = df_no, weights = Innings,  
  family = binomial(link = logit),  
)
```

Statistical model

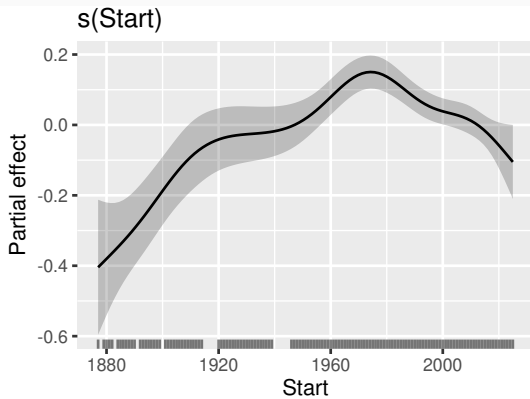
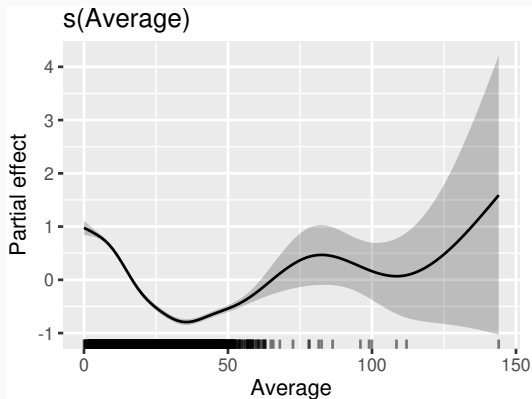
NotOuts ~ Binomial(Innings, q)
 $\log(q/(1 - q)) = g_1(\text{Average}) + g_2(\text{Start})$

Cricket not-outs

Statistical model

$$\text{NotOuts} \sim \text{Binomial}(\text{Innings}, q)$$
$$\log(q/(1 - q)) = g_1(\text{Average}) + g_2(\text{Start})$$

```
fit_notouts <- mgcv::gam(  
  prop_no ~ s(Average) + s(Start),  
  data = df_no, weights = Innings,  
  family = binomial(link = logit),  
)
```



Cricket not-outs

```
augment(fit_notouts, data = df_no) |>  
  mutate(p = surprisals_prob(fit_notouts, approximation = "gpd")) |>  
  select(Player:End, Innings, NotOuts, Average, p) |>  
  arrange(p)
```

A tibble: 3,144 x 8

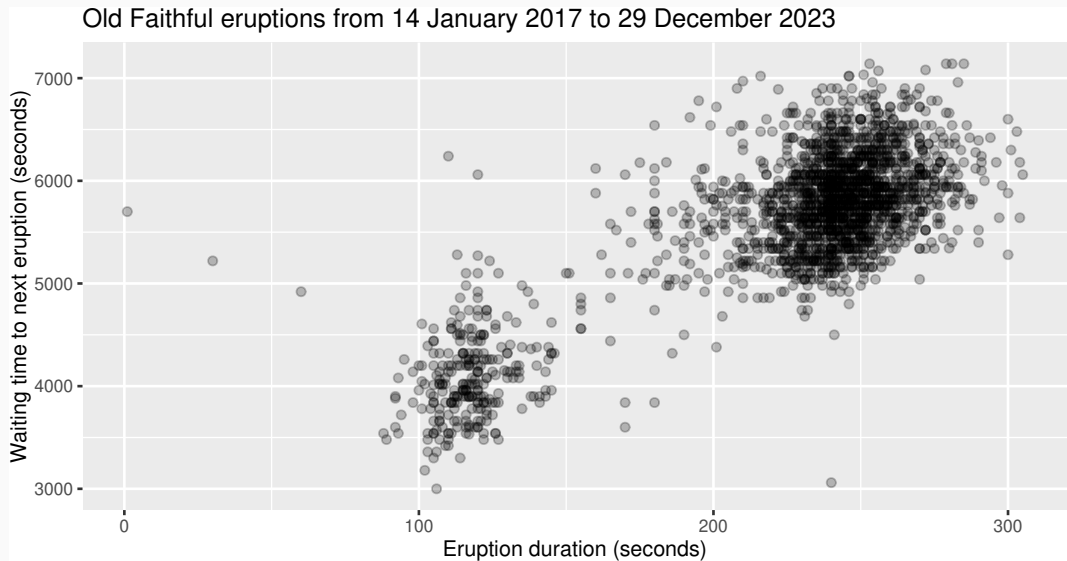
	Player <chr>	Country <chr>	Start <int>	End <int>	Innings <int>	NotOuts <int>	Average <dbl>	p <dbl>
1	TA Boult	New Zealand	2011	2022	94	46	15.8	0.000180
2	SM Pollock	South Africa	1995	2008	156	39	32.3	0.000452
3	JM Anderson	England	2003	2024	265	114	8.96	0.000621
4	JR Hazlewood	Australia	2014	2025	96	47	11.5	0.000812
5	A D'Souza	Pakistan	1959	1962	10	8	38	0.00148
6	Tauseef Ahmed	Pakistan	1980	1993	38	20	17.7	0.00175
7	FJ Cameron	New Zealand	1961	1965	30	20	11.6	0.00231
8	EJ Chatfield	New Zealand	1975	1989	54	33	8.57	0.00236
9	RR Jumadeen	West Indies	1972	1979	14	10	21	0.00257
10	W Rhodes	England	1899	1930	98	21	30.2	0.00286

i 3,134 more rows

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Old Faithful eruptions



Old Faithful eruptions

```
oldfaithful
```

```
# A tibble: 2,097 x 4
```

	time	recorded_duration	duration	waiting
	<dtm>	<chr>	<dbl>	<dbl>
1	2017-01-14 00:06:00	3m 16s	196	5940
2	2017-01-26 14:27:00	~4m	240	5820
3	2017-01-27 23:57:00	2m 1s	121	3900
4	2017-01-30 15:09:00	~4m	240	5280
5	2017-01-31 13:27:00	~3.5m	210	5580
6	2017-01-31 15:00:00	~4m	240	5760
7	2017-02-03 23:13:00	3m 25s	205	5160
8	2017-02-04 22:14:00	3m 34s	214	5400
9	2017-02-05 17:19:00	4m 0s	240	6060
10	2017-02-05 19:00:00	4m 2s	242	6060

```
# i 2,087 more rows
```

Downloaded from geysertimes.org

Old Faithful eruptions

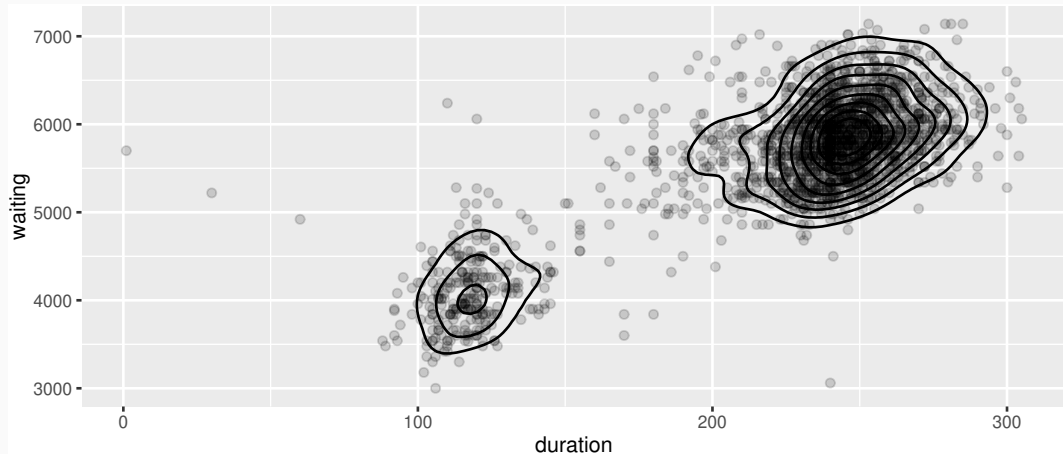
```
oldfaithful$p <- surprisals_prob(  
  oldfaithful[, c("duration", "waiting")],  
  loo = TRUE,  
  approximation = "gpd"  
)  
oldfaithful |> arrange(p)
```

A tibble: 2,097 x 5

	time	recorded_duration	duration	waiting	p
	<dtm>	<chr>	<dbl>	<dbl>	<dbl>
1	2018-04-25 19:08:00	1s	1	5700	0
2	2020-06-01 21:04:00	2 minutes	120	6060	0
3	2020-09-04 01:38:00	>1m 50s	110	6240	0
4	2022-12-03 16:20:00	~4m	240	3060	0
5	2022-12-07 17:19:00	~4 30s	30	5220	0
6	2023-07-04 12:03:00	~1 minute 55ish seconds	60	4920	0
7	2023-07-23 21:57:00	~3 minutes 40ish seconds	180	6540	0.000817
8	2022-09-08 05:13:00	5 min	300	5280	0.000853
9	2021-01-22 18:35:00	2m50s	170	3600	0.000878

Old Faithful eruptions

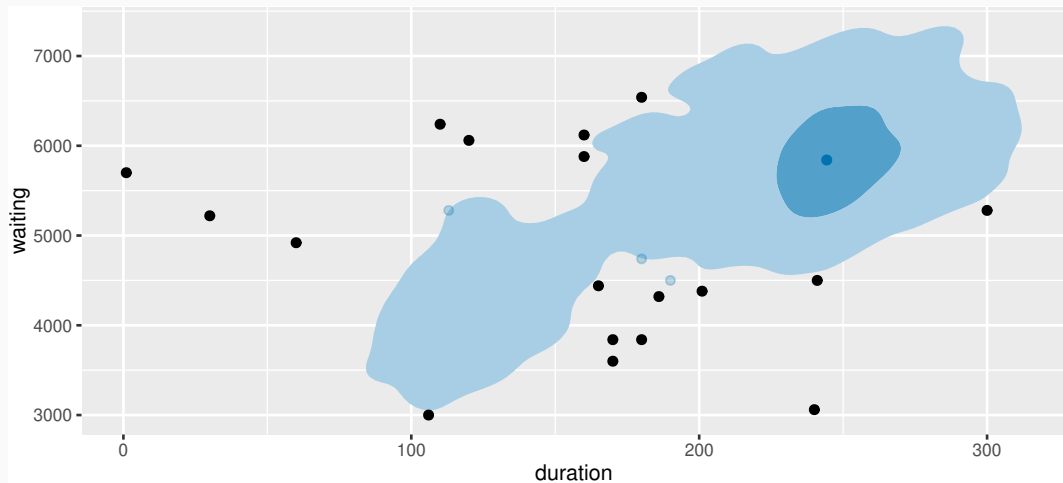
```
oldfaithful[, c("duration", "waiting")] |>  
  dist_kde() |>  
  gg_density(show_points = TRUE)
```



Old Faithful eruptions

$\alpha = 0.005$

```
oldfaithful |>  
  gg_hdrboxplot(duration, waiting)
```



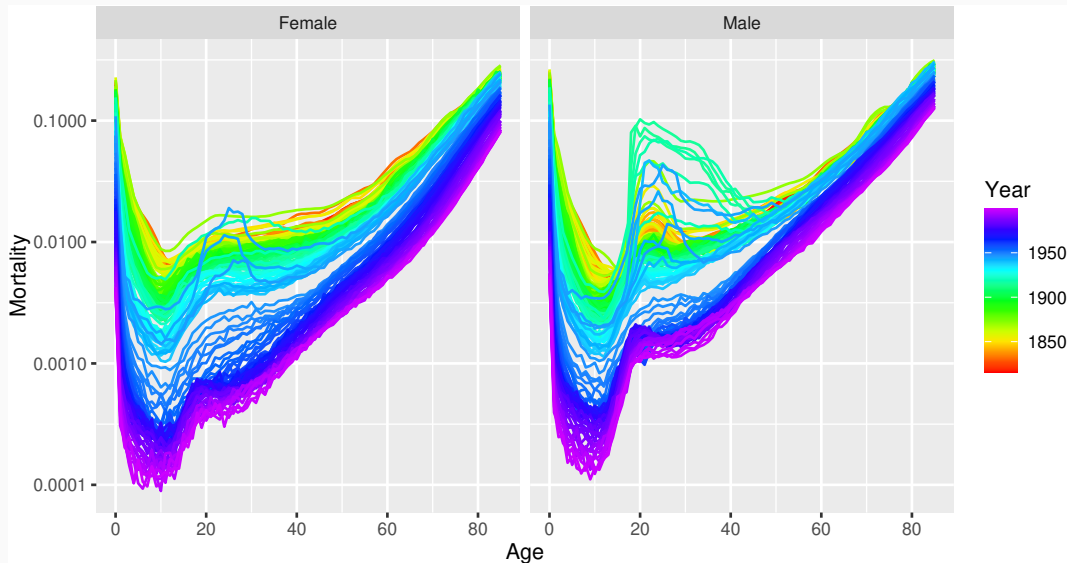
Outline

- 1 What is an anomaly?
- 2 Wine quality and prices
- 3 Approximate extreme surprisal probabilities
- 4 Cricket not-outs
- 5 Old Faithful eruptions
- 6 French mortality rates

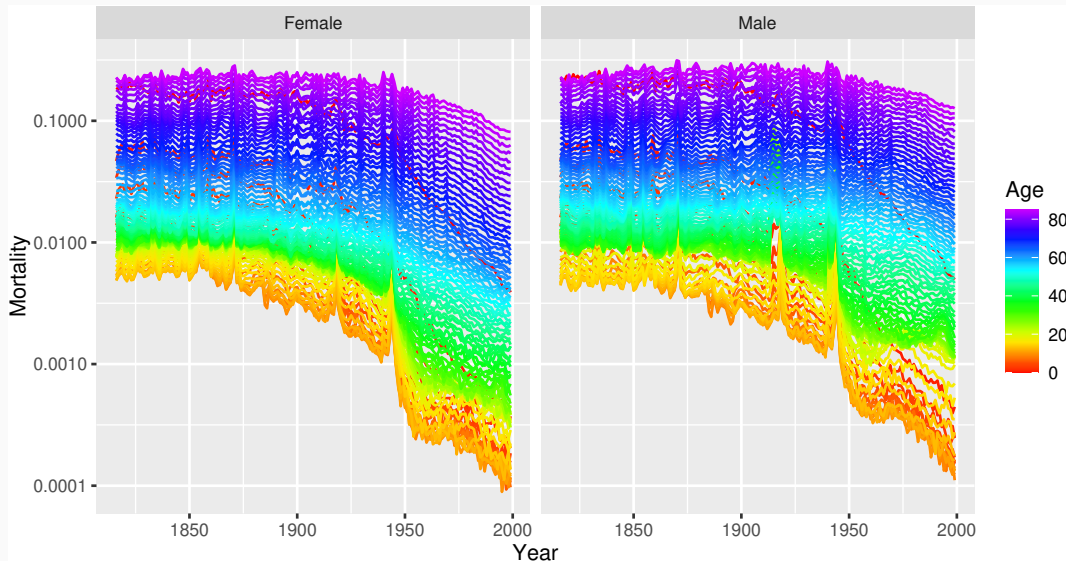
French mortality rates

```
# A tibble: 31,648 x 4
  Year   Age Sex   Mortality
  <int> <int> <chr>     <dbl>
1  1816     0 Female  0.187
2  1816     1 Female  0.0467
3  1816     2 Female  0.0339
4  1816     3 Female  0.0229
5  1816     4 Female  0.0160
6  1816     5 Female  0.0138
7  1816     6 Female  0.0121
8  1816     7 Female  0.0104
9  1816     8 Female  0.00891
10 1816     9 Female  0.00760
# i 31,638 more rows
```

French mortality rates



French mortality rates



French mortality rates

Hampel identifier

$$m_t = \text{median}(y_{t-h}, \dots, y_{t+h})$$

$$a_t = \text{median}(|y_{t-h} - m_t|, \dots, |y_{t+h} - m_t|)$$

$$e_t = |y_t - m_t| / a_t$$

- Anomaly if $e_t > k/0.6745$

French mortality rates

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- Implicit model: $y_t \sim N(m_t, \sigma_t^2)$

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- Equivalent to surprisal probability threshold of $\alpha = 0.0027$ for $k = 3$

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- Anomaly if $e_t > k/0.6745$
- Implicit model: $y_t \sim N(m_t, \sigma_t^2)$
- Anomaly if y_t is more than $k\sigma_t$ from m_t
- Equivalent to surprisal probability threshold of $\alpha = 0.0027$ for $k = 3$
- e_t values are not independent

French mortality rates

```
fr_anomalies <- fr_mortality |>
  group_by(Age, Sex) |>
  mutate(
    hampel = hampel_anomalies(
      Mortality,
      bandwidth = 7,
      alpha = 0.01,
      approximation = "gpd"
    )
  ) |>
  ungroup() |>
  filter(hampel)
```

French mortality rates

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fr_anomalies <- fr_mortality |>
  group_by(Age, Sex) |>
  mutate(
    hampel = hampel_anomalies(
      Mortality,
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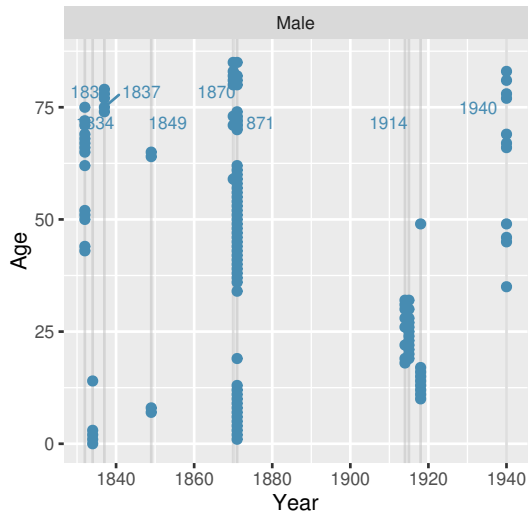
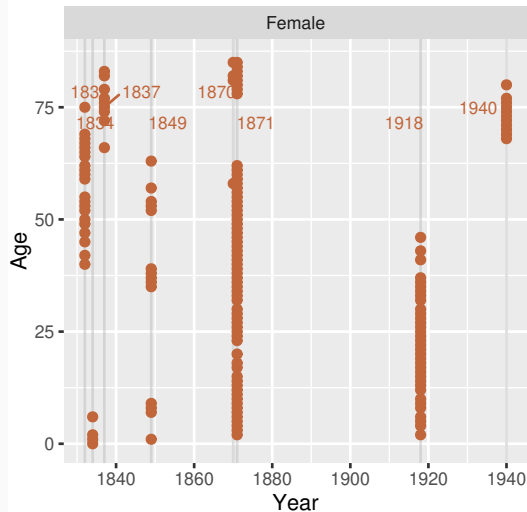
```
fr_anomalies |>
  count(Year, Sex) |>
  filter(n >= 3)
```

	Year	Sex	n
1	1832	Female	21
2	1832	Male	14
3	1834	Female	4
4	1834	Male	5
5	1837	Female	9
6	1837	Male	5
7	1849	Female	14
8	1849	Male	4
9	1870	Female	4
10	1870	Male	8
11	1871	Female	64
12	1871	Male	51
13	1914	Male	8
14	1915	Male	12
15	1918	Female	35
16	1918	Male	9
17	1940	Female	11
18	1940	Male	11

French mortality rates

$k = 6$

French mortality anomalies



French mortality rates

- 1832, 1849: Cholera outbreaks
- 1870: Franco–Prussian war
- 1871: Repression of the ‘Commune de Paris’
- 1914–1918: World War I
- 1918: Spanish flu outbreak
- 1940–1944: World War II

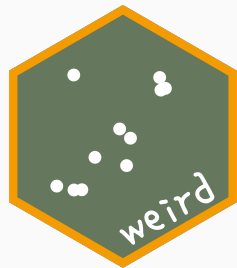
French mortality rates

- 1832, 1849: Cholera outbreaks
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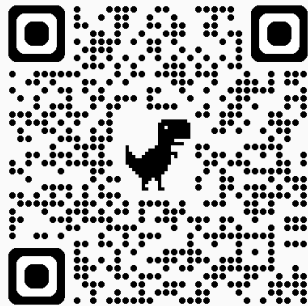
But what happened to:

- infant mortality in 1834?
- elder mortality in 1837?

Find out more



- R package **weird** available on CRAN
- *That's weird* book in preparation at OTexts.com/weird
- Slides, links and related papers: robjhyndman.com/ihaka2026



Coauthors



Sevvandi
Kandanaarachchi
CSIRO



Kate
Turner
ANU



David
Frazier
Monash U